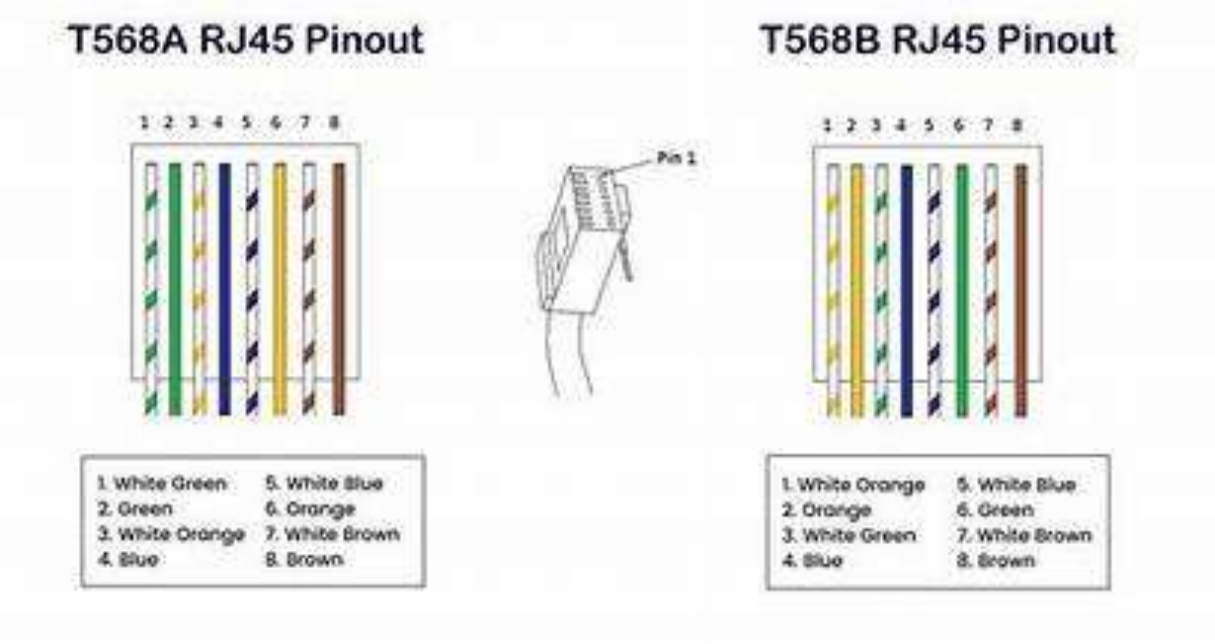
TASK-1 Draw a simple diagram showing the pin layout of an RJ-45 connector and label the color codes for the T568A and T568B Ethernet wiring standards.



TASK-2 Create a table comparing the key specifications of Ethernet standards 10BASE-T, 100BASE-TX, and 1000BASE-T, including speed, cable type, maximum cable length, and connector type.

Specification	10BASE-T	100BASE-TX	1000BASE-T
Speed	10 Mbps	100 Mbps	1000 Mbps (1 Gbps)
Cable type	Cat3 or higher	Cat5 or higher	Cat5e or higher
Maximum cable length	100 metres	100 metres	100 metres
Connector type	RJ-45	RJ-45	RJ-45

TASK-3 Given a scenario where you need to connect two laptops directly using Ethernet, explain which cable (straight-through or crossover) you would use and why.Hint: Think about how transmit and receive pairs are wired in each cable type.

A crossover cable is traditionally used to connect two laptops directly using Ethernet.

- Reason: Both laptops transmit and receive data on the same pin pairs.
- Crossover cable: Swaps the transmit and receive pairs so that one laptop's transmit signals reach the other laptop's receive pins.
- Straight-through cable: Uses the same wiring order at both ends and is traditionally used to connect different devices, such as a laptop to a switch.

TASK-4 List three real-world devices in your home or college that use Ethernet connections, and for each, specify the likely Ethernet standard (e.g., 100BASE-TX, 1000BASE-T) they support.

1 - Wi-Fi Router - 1000BASE-T

2 - Desktop Computer in College Lab - 1000BASE-T

3 - Network Printer - 100BASE-TX